

-----AI-BASED-----

STRESS & OVERTHINKING SUPPORT SYSTEM



A Smart Solution for Mental Wellness
And Emotional Balance



Mood
Tracking



Stress
Analysis



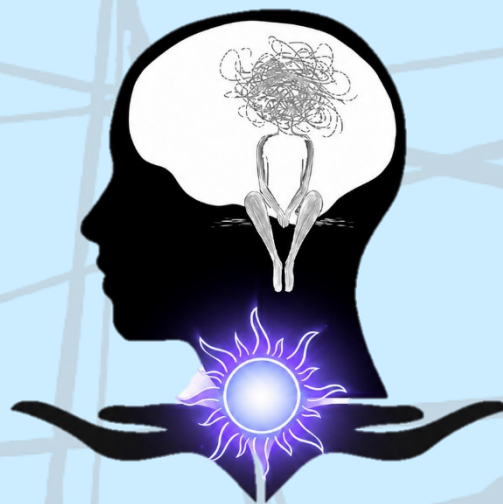
Journal &
Sentiment
Analysis



Personalised
Wellness
Recommendations



Analytics &
PDF Report

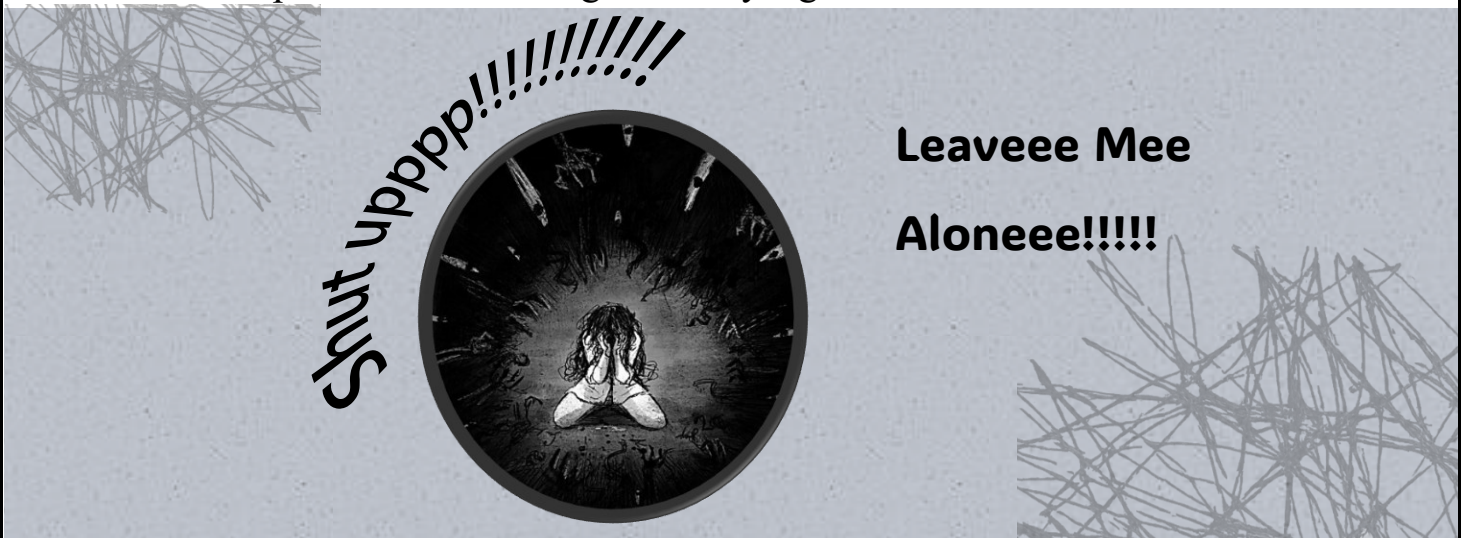


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ABSTRACT

The Mental Wellness System is an AI-assisted emotional wellness support application designed to help users monitor and improve their mental well-being through daily mood tracking and emotional analysis. In today's fast-paced environment, stress, anxiety, overthinking, and emotional imbalance have become common problems, especially among students and working professionals. Many individuals struggle to understand their emotional patterns and often ignore early signs of mental stress.



This project provides a simple and effective solution by allowing users to record their mood score, sleep hours, stress level, energy level, and daily thoughts through a journal system. The system analyses user input and determines emotional condition based on mood patterns and stress indicators. It also identifies signs of overthinking or negative emotional behaviour through journal sentiment analysis.

The application is developed using Python for core programming and MySQL for database management. It also integrates data visualization through graphs to track emotional trends and generate wellness insights. The system recommends personalised wellness activities such as meditation, breathing exercises, journaling, and relaxation techniques based on user mood levels.

Additionally, the project uses Data Structures and Algorithms (DSA) concepts such as QuickSort for stress ranking, Binary Search for date-wise log search, and Hash Maps for emotional keyword analysis, making the system more efficient and technically strong.

The main objective of this project is to create an accessible, user-friendly, and intelligent wellness companion that encourages emotional self-awareness and healthy mental habits. This system is not intended to replace professional mental healthcare but to serve as a supportive emotional tracking tool for everyday life.

INTRODUCTION

Mental health is one of the most important aspects of overall well-being, yet it is often neglected in daily life. Increasing academic pressure, workload, social expectations, and personal challenges can lead to **stress, anxiety, emotional instability, and overthinking**. Many people experience emotional struggles but fail to identify unhealthy mental patterns at an early stage.



Traditional mental health support systems often require professional consultation, which may not always be accessible, affordable, or comfortable for everyone. Therefore, there is a growing need for an easy-to-use digital platform that helps individuals regularly monitor and understand their emotional condition.

The **Mental Wellness System** is designed as a personal emotional support and tracking platform that allows users to maintain a daily record of their mental well-being. The system enables users to log mood scores, sleep hours, stress levels, energy levels, and journal entries to understand emotional trends over time.

Using **sentiment analysis and emotional keyword detection**, the system identifies whether a user appears emotionally stable or emotionally unstable and provides personalised wellness recommendations accordingly. It also generates mood analytics through graphs and allows users to create PDF reports for progress tracking.

The project combines concepts of **Python programming, MySQL database management, Data Structures & Algorithms (DSA), data visualization, and emotional analysis** into a practical real-world application. The purpose of this system is to promote mental wellness awareness and encourage users to build healthy emotional habits through consistent self-monitoring.

This project serves as a technology-driven approach toward emotional wellness and demonstrates practical implementation skills in software development, database systems, and problem-solving.



PROBLEM STATEMENT

In modern society, stress, anxiety, overthinking, emotional burnout, and mental fatigue have become increasingly common due to academic pressure, work responsibilities, social expectations, and personal challenges. However, many individuals fail to monitor their emotional condition regularly and often ignore early warning signs of mental stress.

Most existing mental health solutions are either expensive, complex, or require professional consultation, making them inaccessible for many people. Additionally, many individuals are uncomfortable discussing their emotional condition openly or are unaware of their changing mental patterns over time.

There is a lack of a simple, accessible, and intelligent system that can help users monitor their daily emotional health, analyse behavioural patterns, and provide personalised wellness suggestions.

Therefore, there is a need for a system that can:

- Monitor daily mood and emotional condition
- Detect stress and overthinking patterns
- Analyse journal entries for emotional insights
- Provide personalised wellness recommendations
- Track emotional trends through graphs and reports
- Help users identify emotionally unstable periods early

The **Mental Wellness System** aims to address these issues by providing an easy-to-use platform where users can track emotions, analyse mood trends, and receive supportive wellness guidance. The system promotes emotional self-awareness and encourages users to build healthier mental habits.



OBJECTIVES OF THE PROJECT

The main objectives of the **Mental Wellness System** are:

1. Daily Mood Tracking

To allow users to record daily emotional data such as mood score, stress level, sleep hours, energy level, and personal feelings.

2. Emotional Analysis

To analyse user emotions and determine whether the user appears emotionally stable or emotionally unstable based on mood patterns and journal content.

3. Overthinking Detection

To identify negative emotional keywords and repetitive stress-related thoughts through journal analysis.

4. Personalised Wellness Recommendations

To suggest wellness activities such as breathing exercises, meditation, gratitude journaling, relaxation activities, and stretching based on emotional condition.

5. Data Visualization

To generate mood trend graphs and stress analytics to help users understand emotional changes over time.

6. High Stress Detection

To identify and display days with the highest stress levels for better emotional awareness.

7. Report Generation

To generate downloadable PDF reports containing emotional wellness data and analytics.

8. Efficient Data Handling Using DSA

To improve system performance using **QuickSort, Binary Search, and Hash Maps** for efficient searching, sorting, and emotional pattern analysis.

9. Database Management

To securely store user data, mood logs, and journal entries using **MySQL database integration**.

FEATURES OF THE SYSTEM

The **Mental Wellness System** includes several intelligent features that help users track, analyse, and improve their emotional well-being. The system is designed to provide a simple and user-friendly experience while helping users monitor mental wellness effectively.

1. User Registration & Login System

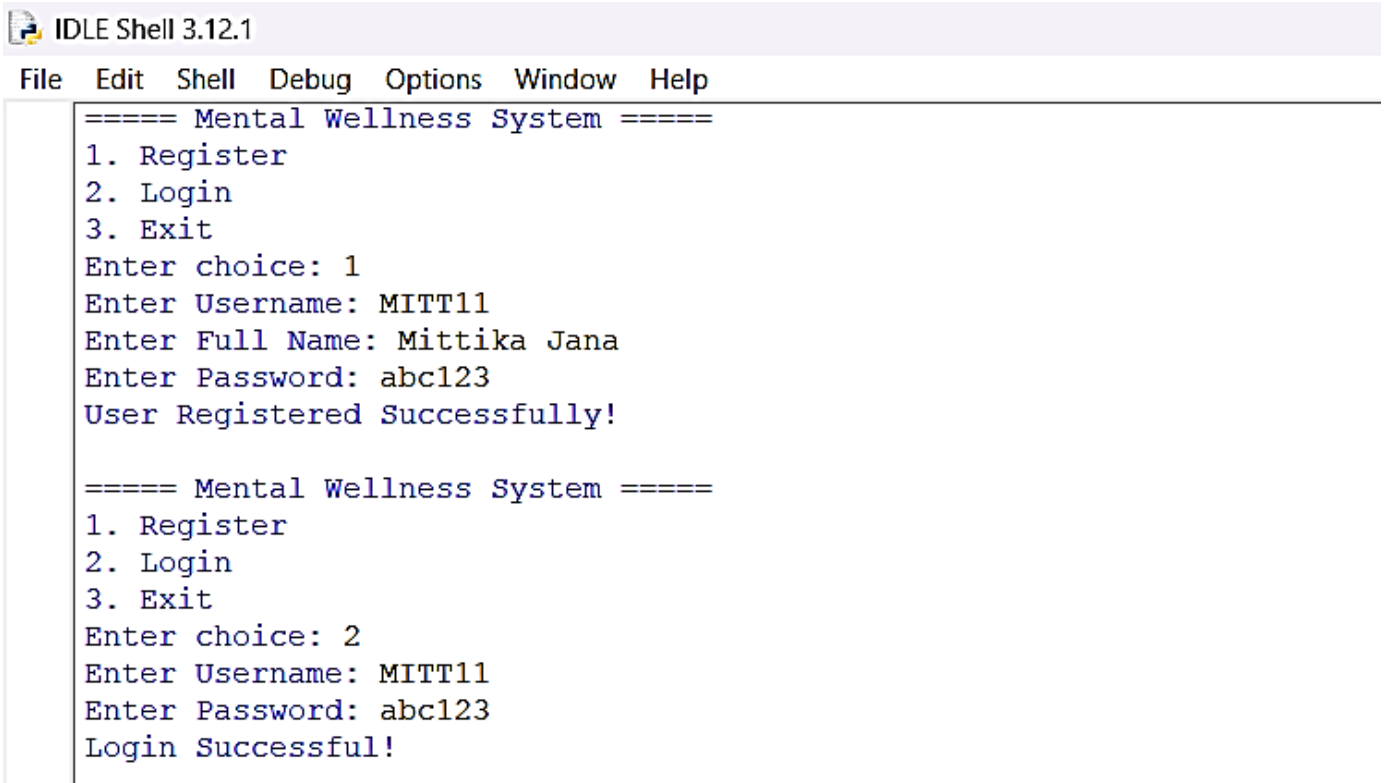
The system provides a secure **user registration and login module** where users can create an account and access their personal wellness dashboard.

Functions

- New user registration
- User authentication
- Secure login access
- Personalized dashboard for each user

Purpose

This feature ensures that each user's emotional records and wellness data remain organised and private.



```
===== Mental Wellness System =====
1. Register
2. Login
3. Exit
Enter choice: 1
Enter Username: MITT11
Enter Full Name: Mittika Jana
Enter Password: abc123
User Registered Successfully!

===== Mental Wellness System =====
1. Register
2. Login
3. Exit
Enter choice: 2
Enter Username: MITT11
Enter Password: abc123
Login Successful!
```

2. Mood Tracking Module

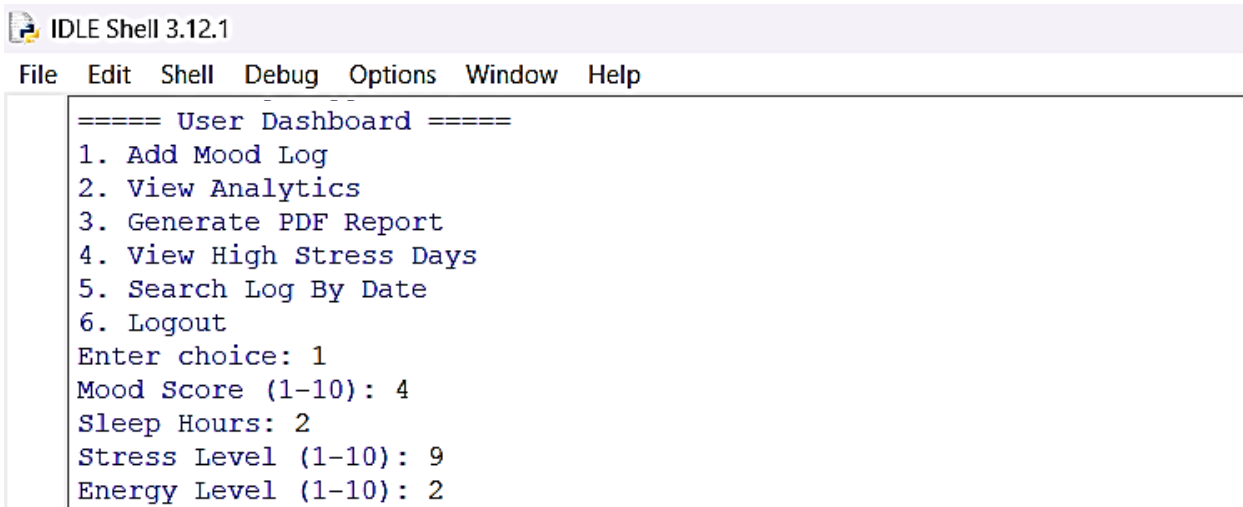
The mood tracking module allows users to record their daily emotional condition by entering important wellness-related information.

Inputs Taken

- Mood Score (1–10)
- Sleep Hours
- Stress Level (1–10)
- Energy Level (1–10)
- Daily Thoughts/Journal Entry

Purpose

This helps users maintain a daily emotional record and understand how mood changes over time.



```
IDLE Shell 3.12.1
File Edit Shell Debug Options Window Help
===== User Dashboard =====
1. Add Mood Log
2. View Analytics
3. Generate PDF Report
4. View High Stress Days
5. Search Log By Date
6. Logout
Enter choice: 1
Mood Score (1-10): 4
Sleep Hours: 2
Stress Level (1-10): 9
Energy Level (1-10): 2
```

3. Journal & Sentiment Analysis

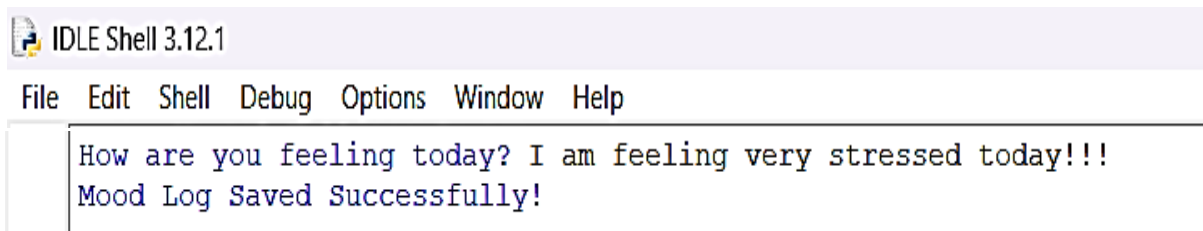
The system allows users to write about how they are feeling through a journal entry system.

The journal is analysed to:

- Detect negative emotional keywords
- Identify overthinking patterns
- Analyse emotional behaviour

Purpose

This helps users understand emotional triggers and mental patterns.



```
IDLE Shell 3.12.1
File Edit Shell Debug Options Window Help
How are you feeling today? I am feeling very stressed today!!!
Mood Log Saved Successfully!
```

4. Mental Analysis System

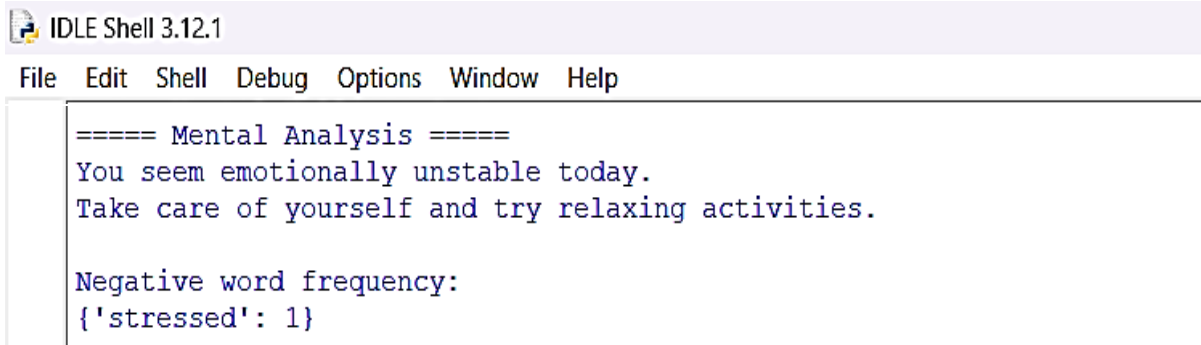
Based on user mood score, stress level, energy level, and journal sentiment, the system determines emotional condition.

The system displays whether the user appears:

- Emotionally Stable
- OR
- Emotionally Unstable

Purpose

To provide users with awareness of their emotional state and encourage self-care.



```
IDLE Shell 3.12.1
File Edit Shell Debug Options Window Help
===== Mental Analysis =====
You seem emotionally unstable today.
Take care of yourself and try relaxing activities.

Negative word frequency:
{'stressed': 1}
```

5. Personalised Wellness Recommendation System

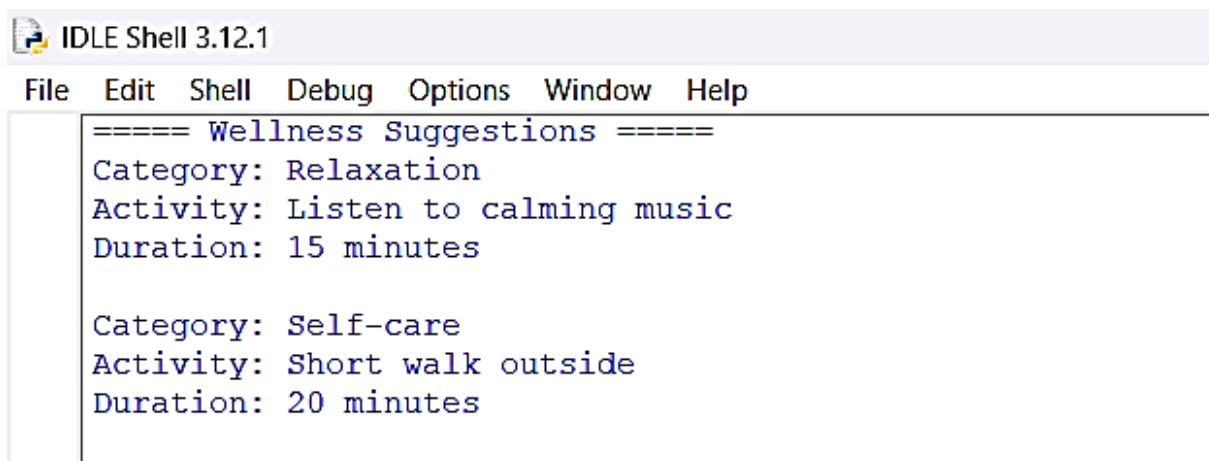
The system provides wellness suggestions according to emotional condition.

Examples of Recommendations

- Deep breathing exercises
- Meditation
- Gratitude journaling
- Walking/stretching
- Relaxation activities

Purpose

To help users reduce stress and improve emotional balance.



```
IDLE Shell 3.12.1
File Edit Shell Debug Options Window Help
===== Wellness Suggestions =====
Category: Relaxation
Activity: Listen to calming music
Duration: 15 minutes

Category: Self-care
Activity: Short walk outside
Duration: 20 minutes
```


6. Mood Analytics & Graph Visualization

The system generates mood trend graphs using analytics tools.

Graph Features

- Mood trend analysis
- Emotional pattern tracking
- Stress-level monitoring

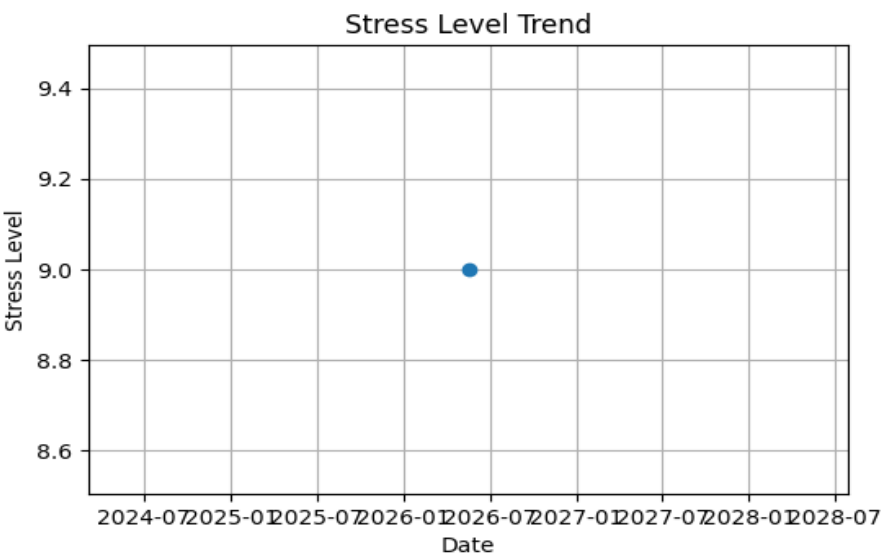
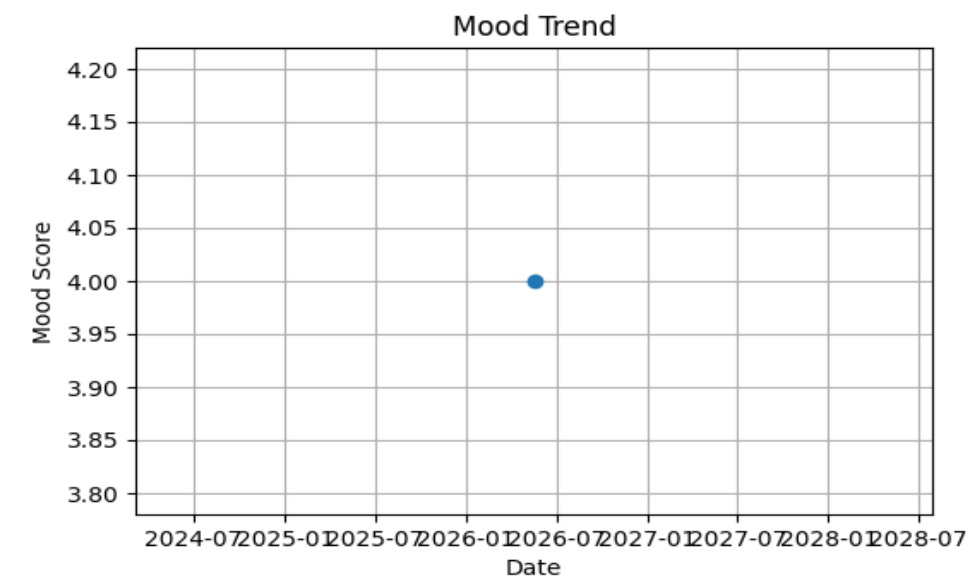
Purpose

To help users visually understand emotional progress.

📄 IDLE Shell 3.12.1

File Edit Shell Debug Options Window Help

```
===== User Dashboard =====
1. Add Mood Log
2. View Analytics
3. Generate PDF Report
4. View High Stress Days
5. Search Log By Date
6. Logout
Enter choice: 2
```

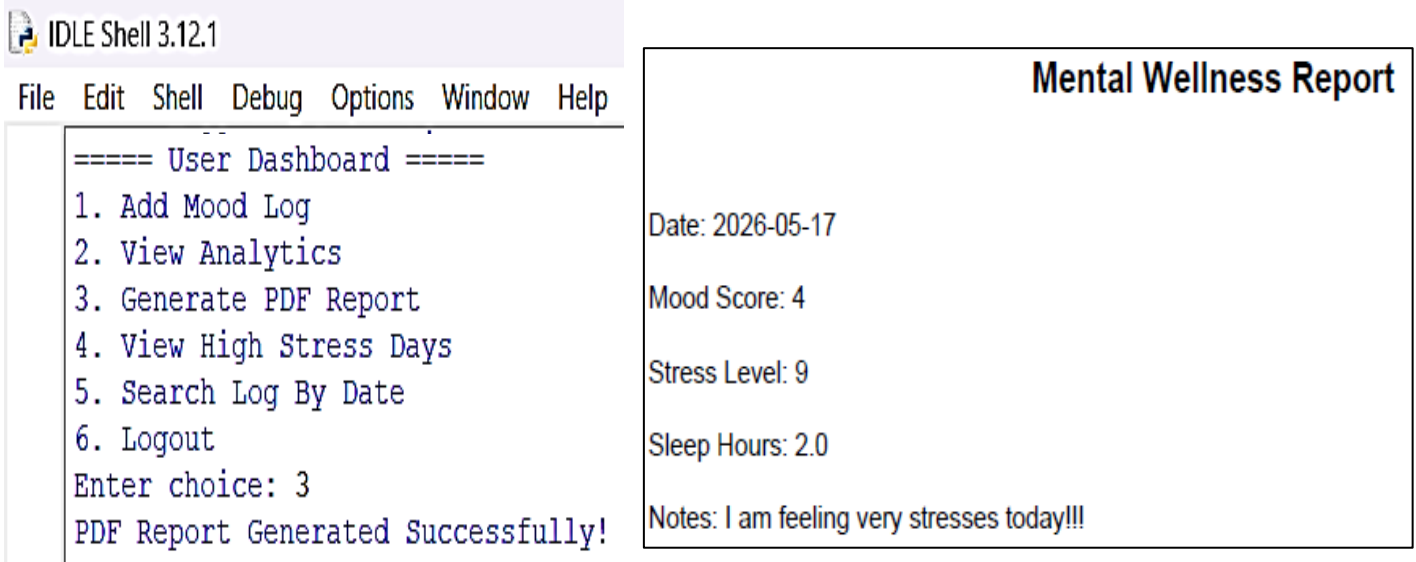


7. PDF Report Generation

The system generates a wellness report in PDF format containing user mood information and analysis.

Purpose

Allows users to maintain emotional progress records professionally.



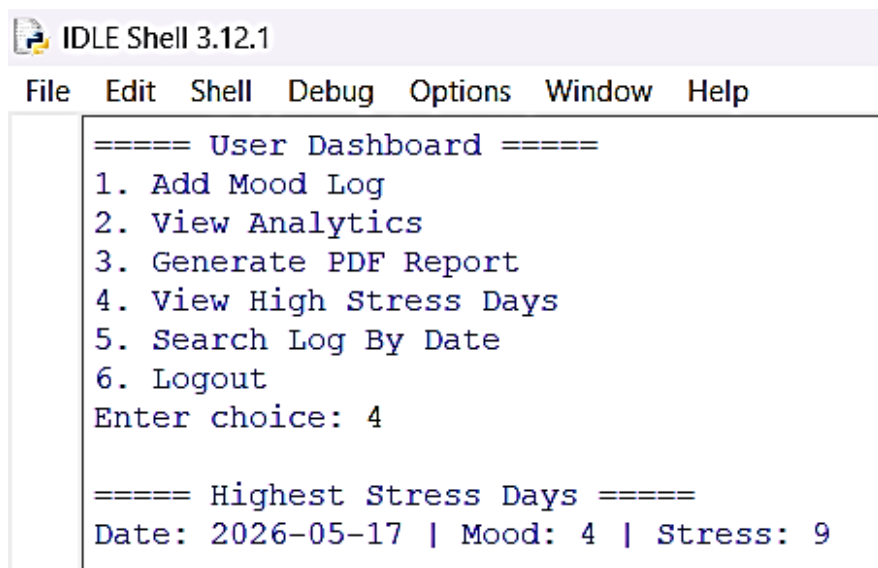
The screenshot shows the IDLE Shell 3.12.1 interface. On the left, a menu titled "User Dashboard" lists six options: 1. Add Mood Log, 2. View Analytics, 3. Generate PDF Report, 4. View High Stress Days, 5. Search Log By Date, and 6. Logout. Below the menu, it says "Enter choice: 3" and "PDF Report Generated Successfully!". On the right, a box titled "Mental Wellness Report" displays the following information: Date: 2026-05-17, Mood Score: 4, Stress Level: 9, Sleep Hours: 2.0, and Notes: I am feeling very stresses today!!!

8. High Stress Day Detection

The system identifies and displays days with the highest stress level using sorting techniques.

Purpose

Helps users identify stressful periods and emotional triggers.



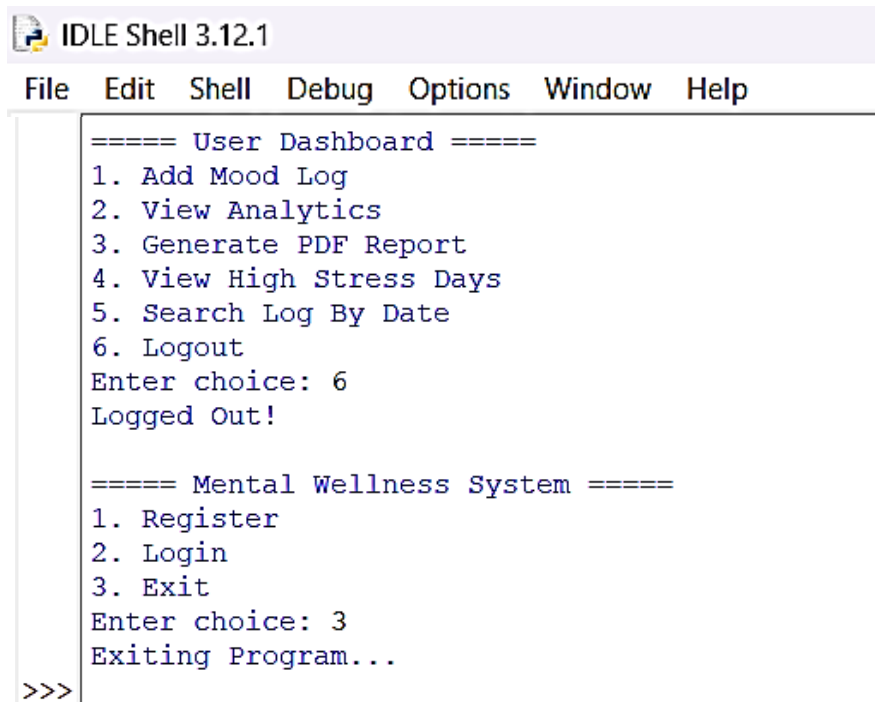
The screenshot shows the IDLE Shell 3.12.1 interface. On the left, a menu titled "User Dashboard" lists six options: 1. Add Mood Log, 2. View Analytics, 3. Generate PDF Report, 4. View High Stress Days, 5. Search Log By Date, and 6. Logout. Below the menu, it says "Enter choice: 4". On the right, a box titled "Highest Stress Days" displays the following information: Date: 2026-05-17 | Mood: 4 | Stress: 9.

9. Search Mood Log by Date

Users can search previously stored emotional logs using dates.

Purpose

Makes data retrieval fast and organised.



```
===== User Dashboard =====
1. Add Mood Log
2. View Analytics
3. Generate PDF Report
4. View High Stress Days
5. Search Log By Date
6. Logout
Enter choice: 6
Logged Out!

===== Mental Wellness System =====
1. Register
2. Login
3. Exit
Enter choice: 3
Exiting Program...
>>>
```

TECHNOLOGY STACK

The **Mental Wellness System** is developed using modern programming tools and technologies to ensure proper functionality, efficient data storage, analytics, and report generation.

TECHNOLOGY	PURPOSES
Python	Core programming language used to build the application logic
MySQL	Database management system used to store user and mood data
MySQL Connector	Connects Python application with MySQL database
Matplotlib	Used for plotting mood analytics and trend graphs
Pandas	Used for handling and analysing mood data
ReportLab	Used for generating PDF wellness reports
DSA (Python)	Used for sorting, searching, and emotional pattern analysis
IDLE	Development environment for coding and testing

DATA STRUCTURES & ALGORITHMS (DSA) USED

To improve efficiency and strengthen the technical implementation of the project, several **Data Structures and Algorithms (DSA)** concepts have been used.

1. QuickSort Algorithm

The **QuickSort algorithm** is used to identify and rank **high-stress days** based on stress levels.

Purpose

- Sort mood logs according to stress level
- Display highest stress days efficiently

Benefit

Provides faster sorting performance for emotional records.

2. Binary Search Algorithm

The **Binary Search algorithm** is used for **date-wise mood log searching**.

Purpose

- Search mood records quickly
- Retrieve emotional logs efficiently

Benefit

Reduces search time compared to linear searching.

3. Hash Map / Dictionary

A **Hash Map (Python Dictionary)** is used for **negative emotional keyword detection** in journal entries.

Purpose

- Count repeated emotional words
- Detect signs of overthinking
- Identify stress-related emotional patterns

Benefit

Enables faster word frequency analysis.

4. List-Based Mood History Tracking

Python Lists are used to maintain mood history and emotional records.

Purpose

- Store mood logs temporarily
- Generate analytics and reports

Benefit

Helps organise emotional data systematically.

CODE IMPLEMENTATION

1. Database Connection (database.py)

```
database.py - D:\MentalWellnessSystem\database.py (3.12.1)
File Edit Format Run Options Window Help
import mysql.connector

def connect_db():
    connection=mysql.connector.connect(
        host="localhost",
        user="root",
        password="cindra@2305",
        database="mental_wellness"
    )

    return connection
```

2. Mood Log Storage (main.py)

```
main.py - D:\MentalWellnessSystem\main.py (3.12.1)
File Edit Format Run Options Window Help

query = """
INSERT INTO mood_logs
(user_id, log_date, mood_score,
sleep_hours, stress_level,
energy_level, notes)
VALUES (%s, %s, %s, %s, %s, %s, %s)
"""

values = (
    user_id,
    today,
    mood_score,
    sleep_hours,
    stress_level,
    energy_level,
    notes
)
```


3. Overthinking Detection (sentiment.py)

```
sentiment.py - D:\MentalWellnessSystem\sentiment.py (3.12.1)
File Edit Format Run Options Window Help
from dsa import HashMap

negative_words = [
    "sad",
    "stress",
    "stressed",
    "anxiety",
    "anxious",
    "worried",
    "depressed",
    "tired",
    "fear",
    "angry",
    "cry",
    "alone",
    "lonely",
    "pain",
    "bad"
]

def detect_overthinking(text):
    words = text.lower().split()
    hashmap = HashMap()
    found_words = []
    for word in words:
        if word in negative_words:
            hashmap.insert(word)
            found_words.append(word)
    negative_count = len(found_words)
    return (
        negative_count,
        hashmap.get_data()
    )
```

4. Hash Map, QuickSort, Binary Search (dsa.py)

```
dsa.py - D:\MentalWellnessSystem\dsa.py (3.12.1)
File Edit Format Run Options Window Help

# HASH MAP FOR NEGATIVE WORD COUNT
class HashMap:
    def __init__(self):
        self.word_count = {}
    def insert(self, word):
        if word in self.word_count:
            self.word_count[word] += 1
        else:
            self.word_count[word] = 1
    def get_data(self):
        return self.word_count

# QUICKSORT FOR STRESS LEVELS
def quick_sort(data):
    if len(data) <= 1:
        return data
    pivot = data[0]
    left = []
    right = []
    for item in data[1:]:
        if item[2] > pivot[2]:
            left.append(item)
        else:
            right.append(item)
    return quick_sort(left) + [pivot] + quick_sort(right)

# BINARY SEARCH FOR DATE
def binary_search(logs, target_date):
    low = 0
    high = len(logs) - 1
    while low <= high:
        mid = (low + high) // 2
        if str(logs[mid][0]) == target_date:
            return logs[mid]
        elif str(logs[mid][0]) < target_date:
            low = mid + 1
        else:
            high = mid - 1
    return None
```

5. Graph Analytics (analytics.py)

analytics.py - D:\MentalWellnessSystem\analytics.py (3.12.1)

File Edit Format Run Options Window Help

```
def generate_graph(user_id):
    conn = connect_db()
    cursor = conn.cursor()

    query = """
    SELECT log_date,
           mood_score,
           stress_level,
           sleep_hours
    FROM mood_logs
    WHERE user_id = %s
    """
    cursor.execute(query, (user_id,))
    rows = cursor.fetchall()
    if not rows:
        print("No mood data found!")
        return

    # Convert to DataFrame
    df = pd.DataFrame(
        rows,
        columns=[
            "log_date",
            "mood_score",
            "stress_level",
            "sleep_hours"
        ]
    )

    # Mood Trend Graph
    plt.figure(figsize=(6, 4))
    plt.plot(
        df["log_date"],
        df["mood_score"],
        marker="o"
    )
    plt.title("Mood Trend")
    plt.xlabel("Date")
    plt.ylabel("Mood Score")
    plt.grid(True)
    plt.show()

    # Stress Trend Graph
    plt.figure(figsize=(6, 4))
    plt.plot(
        df["log_date"],
        df["stress_level"],
        marker="o"
    )
    plt.title("Stress Level Trend")
    plt.xlabel("Date")
    plt.ylabel("Stress Level")
    plt.grid(True)
    plt.show()
    cursor.close()
    conn.close()
```

Database Implementation / SQL Outputs

1. Database Creation

```
mysql> CREATE DATABASE mental_wellness;
Query OK, 1 row affected (0.06 sec)

mysql> USE mental_wellness;
Database changed
```

Purpose: Creates the database for storing wellness records.

2. Users Table Output

```
mysql> select * from users;
+-----+-----+-----+-----+-----+
| user_id | username | full_name | password | created_at |
+-----+-----+-----+-----+-----+
| 1 | MITT11 | Mittika Jana | abc123 | 2026-05-17 16:46:32 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Purpose: Stores registered user details.

3. Mood Logs Output

```
mysql> select * from mood_logs;
+-----+-----+-----+-----+-----+-----+-----+
| log_id | user_id | log_date | mood_score | sleep_hours | stress_level | energy_level |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | 1 | 2026-05-17 | 4 | 2 | 9 | 2 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
+-----+
| notes |
+-----+
| I am feeling very stressed today!!! |
+-----+
```

Purpose: Stores daily emotional records.

4. Journal Entries Output

```
mysql> select * from journal_entries;
+-----+-----+-----+-----+
| entry_id | user_id | entry_date | content |
+-----+-----+-----+-----+
| 2 | 1 | 2026-05-17 | I am feeling very stressed today!!! |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
+-----+-----+
| flagged_words | sentiment_score |
+-----+-----+
| {'stressed': 1} | 1 |
+-----+-----+
```

Purpose: Stores user journal/thoughts for emotional analysis.

5. Recommendation Table Output

```
mysql> select * from recommendations;
```

rec_id	mood_min	mood_max	category	activity	duration_min
1	1	3	Stress Relief	Deep breathing exercise	10
2	1	3	Anxiety	5 minute grounding exercise	5
3	4	6	Relaxation	Listen to calming music	15
4	4	6	Self-care	Short walk outside	20
5	7	10	Growth	Write gratitude journal	10
6	7	10	Fitness	Workout or stretching	30

```
6 rows in set (0.00 sec)
```

Purpose: Stores wellness suggestions based on mood range.

FUTURE SCOPE

The **Mental Wellness System** can be further improved with advanced features to make it more intelligent and user-friendly.

1. AI Chatbot Integration

An advanced chatbot can be integrated to provide emotional support and real-time conversations.

2. Real-Time Emotion Prediction

Machine Learning models can be used to predict emotional instability before it becomes severe.

3. Voice Assistant Support

Voice-based interaction can be added for better accessibility.

4. Cloud Database Integration

Cloud storage can be implemented for secure multi-device access.

5. Advanced Recommendation Engine

Personalised therapy recommendations can be generated using AI models.

6. Real-Time Mood Notifications

The system can remind users to log daily mood and maintain consistency.

CONCLUSION

The **Mental Wellness System** successfully demonstrates an intelligent and user-friendly approach toward emotional wellness tracking and self-monitoring. The system enables users to monitor mood, analyse emotional behaviour, detect signs of overthinking, and receive personalised wellness recommendations.

By integrating **Python, MySQL, DSA concepts, analytics, and sentiment analysis**, the system provides a practical solution for maintaining emotional awareness and identifying stress-related patterns. The addition of mood analytics graphs and PDF report generation further improves the usefulness of the system.

The project highlights how technology can be used to support mental wellness in everyday life. Although the system is not a replacement for professional mental healthcare, it serves as a helpful emotional companion that encourages users to develop healthier mental habits and emotional awareness.

Overall, the project demonstrates strong practical implementation skills in **software development, database management, DSA, problem-solving, and data analytics**, making it a valuable real-world project for learning and professional growth.